

Basavaraj Veerayya

Senior Software Engineer

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Place: Bangalore



Summary:

Senior Software Engineer with 4.5+ years of experience in **Python-driven test automation, API validation, and CI/CD integration**. Skilled in designing scalable test frameworks, debugging complex systems, and ensuring reliability through unit, regression, and system testing. Strong expertise in **pytest, REST API** testing with Postman, and database validation using SQL and in-memory databases. Adept at integrating automated pipelines with **Jenkins, GitHub Actions, and GitLab**, improving release quality and coverage reporting. Proven ability to collaborate in Agile environments, align testing strategies with SDLC/V-Model practices, and drive continuous improvement by reducing manual effort through automation

Skills:

Programming:

>> Python
>> C
>> C++
>> SQL
>> YAML
>> Shell Scripting

Testing and Automation:

>> Python Scripting
>> Python Automation
>> Flask, Rest API, Selenium
>> Pytest, Google Test
>> API & Integration Testing
>> Requests, beautifulsoup
>> Postman, System Validation
>> Test Automation
>> CI/CD Actions, Pipeline
>> Hands on Debugging

Data & Machine Learning:

>> Annotation, Data Validation
>> Machine Learning, Statistics
>> Linear & Logistic Regression,
>> Classification, SVM, KNN,
>> Ensembling, Optimization
>> Pandas, Numpy, Sklearn,
>> Data Cleaning, Preprocessing
>> Feature Engineering, Analysis
>> ML Model Validation
>> Evaluation Metrics

Tools:

>> Cloud Platforms:
AWS S3 Bucket, EC2
>> Version Control:
Github Actions,
GitLab CI, Jenkins
Pipeline setup
>> Supporting Tools:
VS Code, matlab
>> Uniflash, CCS
>> Windows, Linux

EXPERIENCE

Senior Engineer:

KPIT Technology

Dec-2024 to Dec 2025

PROJECT: SBR (Seat Belt Reminder)

Goal: To build an ML-based occupant classification system using in-cabin sensor data, optimizing thresholds and training SVM models for accurate adult vs. child detection,

Job Responsibilities:

- Built code, generated binaries, and handled flashing/debugging with CCS while driving system-level hardware testing for **ECU** integration and **UART** reliability.
- Executed **radar point cloud validation** and CFAR tuning to improve occupant classification accuracy.
- Led **Pytest-based automation** in CI/CD pipelines, enforcing unit, regression, and API validation.
- Implemented **REST API testing** with Postman, and Requests delivering complete coverage and reporting.
- Supervised **data workflows** with Python, Pandas, and NumPy, optimizing large-scale datasets and reducing manual effort.
- Guided **ML model development and validation** (SVM, Logistic Regression, KNN, Ensemble) to enhance detection accuracy.
- Owned issue resolution by **fixing hardware/software defects** and coordinated with stakeholders to align testing strategies with STLC/V-Model practices, ensuring reliability and quality.

PROJECT: SOP2 and OLOC

Goal: To detect life presence inside a vehicle and classify occupants as adult or child for enhanced safety features.

Job Responsibilities:

- Automated data handling, preprocessing, and reporting with **Python, Pandas, and NumPy**, improving accuracy and reducing manual effort.
- Conducted **feature engineering and model validation** for occupant classification, applying ML techniques (SVM, Logistic Regression, Decision Tree) in real-time workflows.
- Implemented **Pytest frameworks** in CI/CD pipelines for unit, regression, and API validation testing, including REST API endpoint coverage.
- Designed and executed REST API and UI testing using **Selenium, Postman, Requests, and BeautifulSoup**, with schema validation, authentication checks, automated browser workflows, and detailed reporting.
- Applied **STLC and V-Model practices**, ensuring requirement traceability and alignment between design (HLD/LLD) and testing strategies.
- Performed **hardware testing and system-level validation**, ensuring ECU integration, firmware flashing, and reliable performance.
- Conducted **radar point cloud validation** and optimized detection thresholds through CFAR tuning to improve point cloud and occupant classification accuracy.
- Built code, generated binaries, and performed flashing/debugging with **Code Composer Studio (CCS)**, while supporting **C/C++ unit testing with Google Test** to ensure embedded system reliability.
- Identified and fixed **hardware issues** during ECU installation, **UART communication**, and sensor validation, improving overall system stability.
- Collaborated in Agile teams, driving continuous improvement and transparency through test case creation, execution, and reporting.

Test Engineer(Annotation Engineer Then Annotation Lead) Path Partner Tech Mar 2021 to Jan 2023

PROJECT: DMS & OMS (Driver and Occupation Monitoring System)

Goal: The project aims to alert drivers during drowsiness or distraction, identify drivers, and enable eye-based function control.

Job Responsibilities:

- Led annotation teams and boosted ML data quality by 30% through **Python automation**, preprocessing, and dataset delivery for algorithm development.
- Processed video streams with **OpenCV** (FPS conversion, overlays) and automated image annotation workflows; built GUI tools with **Tkinter/PyQt5** to improve reliability.
- Handled **AWS S3 data** with Python boto3, streamlined dataset distribution, and automated repetitive validation tasks during firmware flashing and CAN data extraction.
- Installed ECU on trucks, verified CAN and internet connectivity, and performed OS boot and updates.
- Executed functionality, regression, GUI, and system testing, including camera calibration, ensuring end-to-end reliability.
- Conducted unit and integration testing using **Pytest, Selenium, Postman, Requests**, and BeautifulSoup; validated **REST APIs**, automated UI workflows, and generated detailed reports.
- Built code, generated binaries, and performed flashing/debugging with **CCS**, while executing complete **system-level testing in SIL environments** and resolving hardware issues during ECU integration.

EDUCATION:

LANGUAGES:

Master of Computer Application
UNIVERSITY OF MYSORE,
VISHWAVIDYANILAYA KARYA
SOUDHA CRAWFORD HALL,
MYSORE - 570005

Kannada, English, Hindi, Telugu

CERTIFICATIONS:

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|-------------------------------------|---------------------------------|----------------------|
| • Python Programming | LCIPL Training Center Bangalore | Oct-2019 to Dec-2020 |
| • Data Science and Machine Learning | LCIPL Training Center Bangalore | Jan-2020 to Oct-2021 |
| • Power BI | LCIPL Training Center Bangalore | Jan-2026 |

HOBBIES:

ACHIEVEMENTS:

- | | |
|-----------------------|--|
| • Playing chess | > Best Employee Award in 2024 A+ Grade |
| • Watching movies | |
| • Trekking and Biking | |

Date:
Place: Bangalore

Signature